AML Four-Level Analytics

Descriptive, Diagnostic, Predictive and Prescriptive Analysis of Financial Transactions

DESCRIBE
What happened?

→

DIAGNOSE
Why?

→

PREDICT
What next?

→

PRESCRIBE
What action?

Dataset	Result
Transactions	500
Laundering-labelled cases	1 (0.20%)
Analytical modelling choice	Isolation Forest for anomaly-based triage
Primary caution	One positive label is insufficient for a supervised laundering classifier

Candidate outcome: explain the business question, evidence, modelling decision, limitation and production action in interview-ready language.

1. Executive Summary

Management question	Evidence from the dataset	Decision
What happened?	500 transactions; median received 1,173.98; 99th percentile 1,844,995,209.45.	Use medians and percentiles because values are severely right-skewed.
Why are values difficult to compare?	75 transactions use different receiving and payment currencies.	Normalize amounts using timestamp-aligned FX rates.
Can a supervised model be trusted?	Only 1 positive case (0.20%).	No. Use anomaly scoring only for investigation prioritization.
Is the labelled case an obvious outlier?	It ranks 305 of 500 by anomaly score.	Add behavioural, KYC, purpose, geography and network features.

Core conclusion: The dataset supports exploratory analytics and a governed review queue. It does not support automatic money-laundering classification.

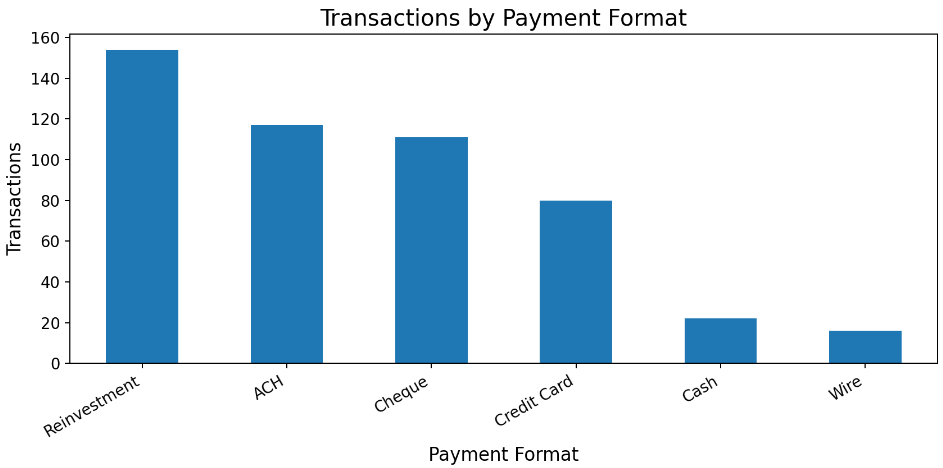
Interview-ready answer

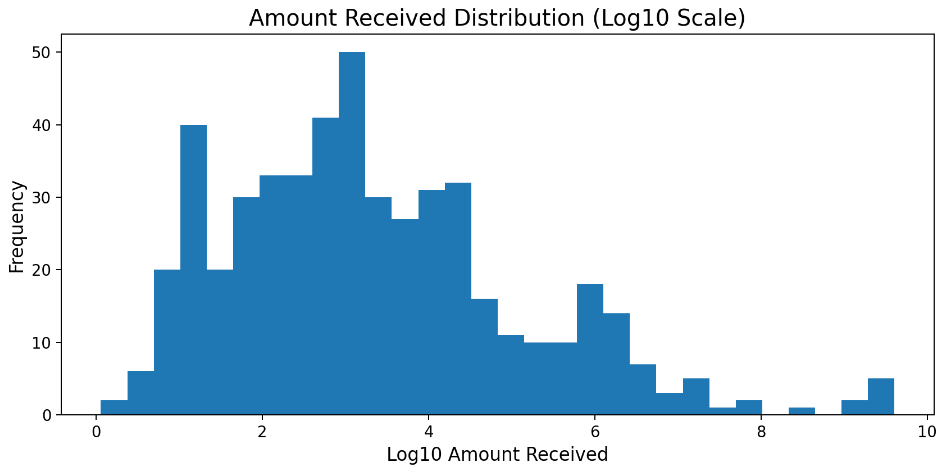
“I first established data quality and class balance. Because the dataset contains only one laundering label, I rejected a misleading supervised classifier and used Isolation Forest to rank unusual transactions. I treated the score as triage evidence, not guilt, and recommended FX normalization, behavioural features, network analytics and investigator feedback before production deployment.”

2. Descriptive Analytics - What Happened?

Metric	Value	Interpretation
Transactions	500	Analytical sample size
Positive labels	1	Extremely rare target class
Label prevalence	0.20%	Severe class imbalance
Median received	1,173.98	Typical transaction
Mean received	44,172,708.68	Inflated by extreme values
95th percentile	3,482,316.81	Upper-tail threshold
99th percentile	1,844,995,209.45	Extreme upper tail
Maximum received	4,016,070,742.00	Largest observed value
Currency mismatches	75	Requires FX normalization

2.1 Distribution and Composition





Reading the charts: payment formats describe transaction composition; the log-scaled amount chart reveals a long right tail that would be hidden on a normal scale.

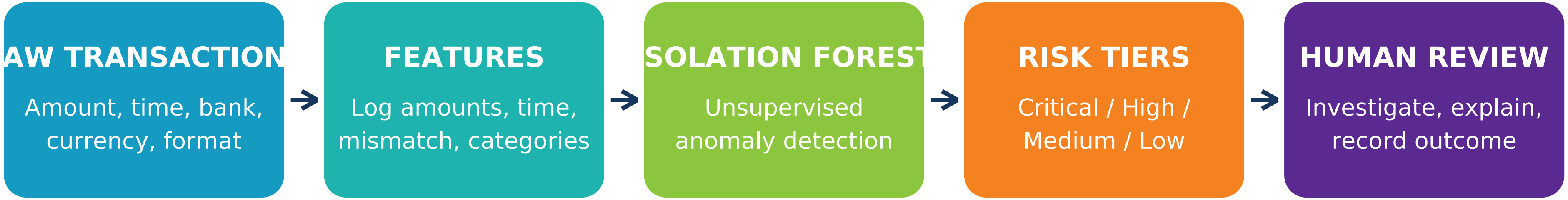
3. Diagnostic Analytics - Why Did It Happen?

Finding	Evidence	Business implication
Extreme amount skew	Mean / median ratio = 37,626.5x	Use robust statistics, logarithms and percentile thresholds.
Class imbalance	1 positive versus 499 negatives	Accuracy would be misleading; prioritize PR-AUC and recall after label growth.
Currency mismatch	75 transactions	Amount difference may reflect exchange rates rather than risk.
No source-bank variation	Unique From Bank values = 1	The feature cannot discriminate risk in this extract.
Label is not an obvious anomaly	Labelled case anomaly rank = 305/500	Transaction context and history are more important than amount alone.

Root-cause reasoning pattern

Observed pattern -> verify data quality -> segment by currency, format, bank and time -> compare labelled and unlabelled cases -> test alternative explanations -> define missing evidence.

4. Predictive Analytics - Which Transactions Need Review?

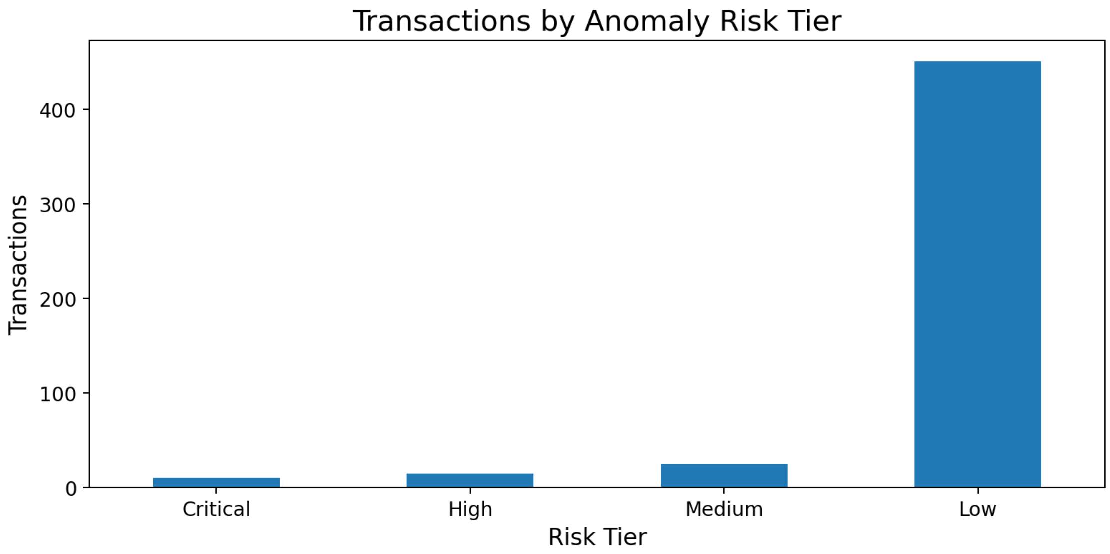


Governance rule: anomaly score is a review-priority signal,
not a probability of money laundering or an automated accusation.

Why Isolation Forest?

It can rank unusual observations without requiring many positive labels. The model isolates rare combinations of amount, time, bank, currency and payment format. It is appropriate for triage, but not for assigning a laundering probability.

4.1 Risk-Tier Distribution



Most transactions remain in the Low tier, while the upper score percentiles create a small, capacity-controlled review queue. Thresholds should be approved according to investigator capacity and the cost of missed cases.

4.2 Highest-Priority Transactions

Rank	Timestamp	To Bank	Amount Received	Recv. Currency	Format	Score	Tier
1	2022-09-03 00:04	4	4,016,070,742.00	Ruble	Cheque	0.6870	Critical
2	2022-09-09 01:41	9	1,673,041.14	Ruble	Credit Card	0.6827	Critical
3	2022-09-03 00:12	4	2,981,124,303.00	Ruble	ACH	0.6826	Critical
4	2022-09-05 14:08	4	4,016,070,742.00	Ruble	Cheque	0.6822	Critical
5	2022-09-06 07:13	4	92,760,199.31	Ruble	Credit Card	0.6769	Critical
6	2022-09-01 23:24	8	29.60	Canadian Dollar	ACH	0.6721	Critical
7	2022-09-02 18:23	4	2,146,543.72	Ruble	Cash	0.6720	Critical
8	2022-09-06 21:12	8	29.60	Canadian Dollar	ACH	0.6706	Critical

5. Prescriptive Analytics - What Should the Business Do?

Priority	Action	Recommendation	Owner	Timeline
P0	Human review queue	Review Critical and High tiers daily.	Compliance Operations	Immediate
P0	FX normalization	Convert values to a common currency using transaction-time FX rates.	Data Engineering	1-2 weeks
P0	Data enrichment	Add KYC risk, geography, purpose, account age and prior alerts.	Compliance + Data	2-4 weeks
P1	Behavioural features	Build 1h/24h/7d velocity, value, round-amount and new-counterparty features.	Data Science	2-4 weeks
P1	Network analytics	Detect cycles, hubs, layering, fan-in/fan-out and rapid pass-through.	Financial Crime Analytics	4-8 weeks
P1	Label improvement	Capture investigator outcomes and SAR decisions.	Compliance Operations	Ongoing
P2	Model governance	Use time validation, explainability, drift monitoring, approvals and audit trails.	Model Risk	6-10 weeks
P2	Threshold optimization	Align alert thresholds with investigator capacity and business cost.	Compliance Leadership	After label growth

6. Production Governance and Interview Revision

Control	What to implement	Interview evidence
Validation	Time-based splits; no future leakage; review-capacity metrics.	Explain why random split may overstate AML performance.
Metrics	Precision, recall, PR-AUC, recall at fixed alerts/day, false-positive cost.	Avoid accuracy for rare-event problems.
Explainability	Show contributing features and transaction context for every alert.	Investigator must understand why a case was prioritized.
Monitoring	Data drift, score drift, alert volume, investigator outcomes and override rate.	Connect model monitoring to operational impact.
Auditability	Version data, features, model, thresholds and case decisions.	Reproduce every alert and decision.

Rapid interview questions

Interview question	Strong answer points
Why not train a classifier?	Only one positive label; accuracy would be meaningless; no reliable generalization.
What does the anomaly score mean?	Relative unusualness for review prioritization, not a laundering probability.
What features are missing?	KYC, purpose, geography, history, velocity, counterparties, network structure and outcomes.

7. Python Revision Snippets

Descriptive statistics

```
# Robust descriptive statistics
summary = df['Amount Received'].agg(['count','median','mean','max'])
p95 = df['Amount Received'].quantile(0.95)
p99 = df['Amount Received'].quantile(0.99)
print(summary, p95, p99)
```

Diagnostic segmentation

```
# Diagnose currency-related differences
diagnostic = (
df.groupby('Currency Mismatch')
.agg(transactions=('Is Laundering','size'),
median_difference=('Absolute Difference','median'),
labels=('Is Laundering','sum'))
)
display(diagnostic)
```

Anomaly scoring

```
model = IsolationForest(
n_estimators=400,
contamination=0.02,
random_state=42
)
model.fit(X)
df['Anomaly Score'] = -model.score_samples(X)
df['Anomaly Rank'] = df['Anomaly Score'].rank(ascending=False)
```

8. One-Page Revision Cheat Sheet

Level	Business question	Method	Key result	Action
Descriptive	What happened?	Counts, medians, percentiles, distributions	500 rows; extreme right skew	Use robust statistics
Diagnostic	Why?	Segmentation, comparison, data-quality checks	Currency mismatch and missing context matter	Normalize FX and enrich data
Predictive	What requires review?	Isolation Forest anomaly ranking	Labelled case ranked 305/500	Use score for triage only
Prescriptive	What should be done?	Prioritized roadmap and governance	Human-in-the-loop production design	Build evidence and feedback loop

Final interview statement

“The analytical value is not in pretending that one positive label can train a fraud model. The value is in proving what the data can support, ranking unusual transactions responsibly, and designing the data, review and governance system needed for a reliable AML capability.”